RAJALAKSHMI ENGINEERING COLLEGE

Kamalesh B 250701308-2025-CSE

K2

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|---|---|
| **Started on** | Wednesday, 5 August 2026, 8:35 AM |
| **State** | Finished |
| **Completed on** | Wednesday, 5 August 2026, 8:49 AM |
| **Time taken** | 14 mins 33 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:** (penalty regime: 0 %)

```
#include<stdio.h>
void function(int n){
    int count=0;
    count++;
    if(n==1){
        count++;
    }
    else{
        count++;
        for(int i=1;i<=n;i++)
        {
            count++;
            for(int j=1;j<=n;j++)
            {
                count++;
                count++;
                break;
            }
            count++;
            count++;
        }
    }
    printf("%d",count);
}
int main(){
    int n;
    if(scanf("%d",&n)==1){
        function(n);
```

```
29      }
30      return 0;
31  }
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 2 | 12 | 12 | ✔ |
| ✔ | 1000 | 5002 | 5002 | ✔ |
| ✔ | 143 | 717 | 717 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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